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Step 11 - Timer Virtualization Concepts

Reference Note

This chapter is based on concepts described in the ARM Generic Timer Architecture documentation.

Previous chapters explained:

System Counter
↓
Physical Counter (CNTPCT)
↓
Physical Timer

AND

System Counter
↓
Virtual Counter (CNTVCT)
↓
Virtual Timer

This chapter explains **why ARM introduced Virtual Counters and Virtual Timers**

Learning Objectives

After completing this chapter, you should understand:

- Why virtualization requires time isolation
- Problems with exposing physical time to guests
- Physical Time vs Guest Time
- Hypervisor responsibilities
- Why ARM uses offset-based virtualization
- Relationship between Physical and Virtual timers

1. What Problem Are We Trying to Solve?

Consider a system running:

Software Layer	Description
Hypervisor	Controls the machine
Guest OS #1	Virtual Machine A
Guest OS #2	Virtual Machine B

All are running on the same hardware.

Question:

Should every guest see the same physical time?

Answer:

Not necessarily.

2. Physical Time

Physical time comes directly from:

Component	Register
System Counter	CNTPCT_EL0

Example:

Register	Value
CNTPCT_EL0	1,000,000,000

This represents:

Actual hardware time

since power-on.

3. Why Physical Time Is Not Ideal For Guests

Suppose:

Item	Value
Hardware Uptime	100 Days
Physical Counter	Large Value

Now a Guest VM boots.

If it reads:

CNTPCT_EL0

it immediately discovers:

System has been running for 100 days

However:

Guest just started.

This is not an ideal virtualization model.

4. Desired Guest View

A newly started guest should see:

Guest Time	Value
System Boot Time	0
Current Runtime	Small Value

Instead of:

Guest Time	Value
System Boot Time	100 Days Ago

ARM solves this by creating:

Virtual Time

5. Physical Time vs Virtual Time

Physical Time	Virtual Time
Actual hardware timeline	Guest-visible timeline
Based on CNTPCT	Based on CNTVCT
Shared by machine	Can differ per guest
Controlled by hardware	Controlled by hypervisor

6. ARM Virtualization Strategy

ARM does NOT create:

Approach	Result
Separate hardware counter per VM	Expensive

Instead ARM uses:

Approach	Result
One System Counter	Shared
Offset per guest	Virtual timelines

This is much simpler.

7. Hypervisor Role

The Hypervisor:

EL2

manages virtual time.

Responsibilities:

Task	Purpose
Configure offset	Create guest timeline
Switch offsets	Change between guests
Maintain isolation	Prevent guest visibility into host time

Think of EL2 as:

Time Manager

for virtual machines.

8. Guest Time Example

Hardware:

Item	Value
Physical Counter	1,000,000

Guest A should see:

0

Guest B should see:

500,000

ARM achieves this without creating new counters.

Instead:

System Counter
+
Offset

produces different guest-visible time values.

9. Timer Virtualization

Timekeeping alone is not enough.

Guests also need timers.

Guest expects:

Set timer
Wait
Receive interrupt

Therefore ARM provides:

Counter	Timer
Virtual Counter	Virtual Timer

Relationship:

Component	Purpose
CNTVCT_EL0	Virtual Time
CNTV_CVAL_EL0	Virtual Deadline
CNTV_TVAL_EL0	Virtual Delay
CNTV_CTL_EL0	Virtual Timer Control

This allows timer events to be based on:

Guest Time

instead of:

Host Time

10. Benefits of Virtual Time

Benefit	Description
Guest Isolation	Guests do not observe host uptime

VM Migration	Guest timeline remains consistent
VM Snapshot/Restore	Guest time can be preserved
Flexible Startup	Guest can start at time zero
Efficient Hardware	No additional counters required

11. Architecture Summary

Component	Purpose
System Counter	Real hardware time
Physical Counter	Direct time view
Virtual Counter	Guest time view
Physical Timer	Physical event generation
Virtual Timer	Guest event generation
Hypervisor	Controls virtual time

Common Misconceptions

Misconception 1

Virtual Time requires another counter.

False.

ARM uses:

One System Counter

plus virtualization logic.

Misconception 2

Virtual Timer needs another clock source.

False.

Both timers ultimately derive time from the same System Counter.

Misconception 3

Physical Time and Virtual Time are unrelated.

False.

Virtual Time is derived from Physical Time.

Key Takeaways

Concept	Key Point
Physical Time	Real hardware time
Virtual Time	Guest-visible time
Hypervisor	Manages guest timelines
Virtual Counter	Derived from Physical Counter
Virtual Timer	Uses Virtual Counter
Goal	Time isolation without extra hardware

Knowledge Check

Q1

Why is physical time not ideal for guest operating systems?

Q2

Who manages virtual time?

Q3

Why doesn't ARM create a separate hardware counter for every VM?

Q4

What timer does a guest typically use?

Q5

What is the goal of timer virtualization?

Answer Key

Question	Answer
Q1	It exposes actual system uptime and host timeline
Q2	Hypervisor (EL2)
Q3	Higher hardware cost and complexity
Q4	Virtual Timer
Q5	Provide isolated guest-visible time